Question		Answer	Marks	AO	Guidance
4	(a)	$4\times3^3-8\times3^2-11\times3-3$ $[=108-72-33-3]=0$	B1 [1]	1.1	arithmetic must be correct and 4 terms must be seen
4	(b)	$(4x^2+4x+1)$ seen $(x-3)(2x+1)^2$ **isw** or $a=2$, $b=1$ **isw**	M1 A1	2.1 1.1	may be embedded in long/synthetic division; allow one coefficient error allow $(x-3)(2x+1)(2x+1)$; mark the final answer; supporting working must be seen for the award of this mark
		Alternative method $a^2=4,\ -3a^2+2ab=-8,$ $b^2-6ab=-11, -3b^2=-3$ $a=2, b=1$ **isw**	M1 A1		for any two equations from equating coefficients; may be embedded in expansion; allow one coefficient error supporting working must be seen for the award of this mark
			[2]		
4	(c)		M1 A1 [2]	1.1 1.1	cubic of correct orientation; curve touches x-axis on correct side of origin for their $-\frac{b}{a}$ and intersects positive x-axis; intercepts at $\left(-\frac{1}{2},0\right)$, $(3,0)$ and $(0,-3)$ labelled or identified on sketch; local minimum in 4th quadrant